

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

1. Introduction and problem statement

1.1 Introduction

Genetic identification is crucial today because it is related to important fields such as healthcare and the criminal justice system. These are some real-world cases in which genetic identification technologies make a difference: knowing information about a patient's genes helps the doctor select medicines that works better; identification of similarity of DNA among family members makes it possible to evaluate risks of inherited diseases, therefore supports early diagnosis and treatment; DNA tests conducted by forensics department facilitates fast identification of criminals and act as important evidence in court. Our project focuses on sample identifiability, a subarea under genetic identification, that determines if pairs or sets of samples come from the same source. Specifically, we are developing a model that evaluates the likelihood of two samples coming from the same patient.

1.2 Problem formulation

Given a collection of blood plasma proteomics samples acquired from multiple patients across multiple time points, the objective is to determine whether any two samples originate from the same individual. Formally:

- Input: A pair of sample vectors (A, B) , where each vector contains the ion count intensities of peptidoforms (modified peptide variants) detected in a single blood plasma mass spectrometry run. Each vector has dimension up to 40,921 (the total number of peptidoforms in the dataset), with entries representing abundance values or N/A for undetected peptidoforms.
- Output: A scalar similarity score $s(A, B)$ indicating the likelihood that samples A and B come from the same patient.
- Objective:
 - Maximize the ability to correctly identify same-patient sample pairs (true positives) while minimizing false matches between samples from different patients (false positives). The model must additionally reject cross-dataset matches, i.e., no sample from the weight-loss cohort (MSV000080596) should be matched to any sample from the external COVID-19 sera cohort (MSV000085507).
 - Assess the worldwide identifiability of each sample: whether the combination of single amino acid polymorphisms (SAAPs) detected in a sample is sufficiently rare in the global population to uniquely identify the source individual with probability less than 1 in 10 billion.

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

1.3 Inputs Definition

Each sample in this project is one blood plasma mass spectrometry run from the plasma weight-loss dataset. In this project, the identifiability task uses 336 samples from 58 patients collected at multiple time points, and the peptidoforms intensities table contains up to 40,921 resolved peptidoforms. Following the dataset instructions, we will use the sample intensity columns and not the corresponding “_unmod” columns when building the input features. Since our task is to decide whether two samples come from the same patient, the model input is either a pair of sample vectors or a feature vector built from that pair.

We will use two main input representations. The first is a SAAP-based representation. For each sample, we build a binary vector that records whether each selected SAAP is detected or not. This representation fits the SAAP similarity model as it compares shared variant patterns between two samples. It also makes biological sense, since SAAPs reflect genetic differences and should stay more stable across time points from the same person. The second is a continuous intensity-based representation built from filtered peptidoform or variant abundance values. For this representation, each sample is described by its measured intensity values after standard preprocessing and feature filtering. This representation fits the random forest, PCA-based similarity, k-NN, and embedding-based models because these methods work with numeric feature vectors and can capture broader proteomic patterns beyond only SAAP presence.

These two representations let us test two kinds of signals in the data. The SAAP-based input focuses on stable identity-related variation, while the intensity-based input captures overall abundance patterns that may also help separate patients. This makes the input design consistent with the models proposed in the later sections.

1.4 Missing Values

Missing values are an important issue in this dataset. The zero values in the dataset tables should be treated as N/A rather than true zero abundance. Our preliminary analysis shows that the peptidoforms data are very sparse: 38,447 of 40,921 features have more than 50% missing values, which indicates that we should not use all raw features directly.

We will first remove features with too many missing values since very sparse features are less reliable and can add noise. For the SAAP-based representation, this is handled naturally through detection: a SAAP is marked as detected or not detected in each sample. For the continuous intensity-based representation, we will first filter and then handle the remaining missing values with simple imputation during preprocessing. We may also include missingness indicators if they appear useful.

It may also be useful to work with subsets of features instead of one full feature set. In particular, we can compare SAAP-only features, broader abundance-based features, and smaller high-confidence subsets. This is reasonable because different models may work better

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

with different kinds of inputs, and feature subsets can reduce noise and improve stability when the number of samples is much smaller than the number of features.

2. Proposed approach

2.1 Proposed models

We propose four models of increasing complexity. The first two serve as interpretable baselines grounded in biological reasoning, and the latter two are learning-based models that may capture nonlinear or combinatorial identity signals.

1. SAAP Jaccard similarity (unsupervised baseline)

For each sample pair (A, B) , compute the Jaccard similarity coefficient between their binary SAAP detection vectors: $J(A, B) = |S_A \text{ AND } S_B| / |S_A \text{ OR } S_B|$, where S_A and S_B are the sets of SAAPs detected in each sample. A high Jaccard score indicates the two samples share a large fraction of their detectable SAAPs, consistent with originating from the same individual.

Rationale: This model is parameter-free, requires no training, and directly tests the biological hypothesis that SAAP detection patterns are individual-specific. It provides a strong interpretable baseline and will reveal whether SAAPs alone carry sufficient identity signals.

2. Weighted SAAP similarity with population-frequency weighting

An extension of Model 1 that weights each SAAP by the inverse of its population allele frequency (from the dbSNP SAAP frequency table). The similarity score becomes a weighted Jaccard or cosine similarity, where the weight for SAAP i is $w_i = -\log_{10}(\text{freq}_i + \epsilon)$, and ϵ is a small constant to handle zero-frequency entries. Matching SAAPs are further weighted by PValue to downweight low-confidence identifications.

Rationale: Rare SAAPs are more informative for identity, since they are less likely to be shared by chance between unrelated individuals. This model directly enables the worldwide identifiability calculation by combining the weights into a cumulative random match probability.

3. Cosine similarity on PCA-reduced intensity features

We compute cosine similarity between sample pairs using the continuous feature representation (Representation B) after dimensionality reduction via PCA. For example, the feature vectors $\vec{a}$

and $\vec{b}$, the similarity is defined as:

$$\text{sim}(\vec{a}, \vec{b}) = \frac{\vec{a} \cdot \vec{b}}{||\vec{a}|| ||\vec{b}||}$$

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

Rationale: This model provides a simple yet strong baseline that captures global similarity in protein abundance profiles. Compared to SAAP-based models, it leverages continuous intensity information rather than binary variant presence. Applying PCA reduces noise and mitigates the curse of dimensionality. This baseline helps determine whether general proteomic patterns alone are sufficient for patient identification.

4. Random forest on pairwise feature differences

For each sample pair (A, B) , compute the feature difference vector $d = |\log_2(A) - \log_2(B)|$ over the shared non-missing features from Representation B. Train a random forest classifier to predict same-patient (label 1) vs. different-patient (label 0) from d .

Rationale: Random forests handle high-dimensional input with implicit feature selection, are robust to irrelevant features, and provide interpretable feature importance scores. The pairwise-difference formulation is symmetric and models the matching task directly. Feature importance output will identify which peptides/proteins contribute most to patient discrimination, guiding the Stage 2 validation analysis.

5. Contrastive metric learning (embedding-based)

Train a shallow neural network (2-3 fully connected layers with ReLU activations) that maps each sample's feature vector (from Representation B, after PCA) to a low-dimensional embedding (e.g., 32-128 dimensions). The network is trained with a contrastive loss function: $L = y \times D^2 + (1 - y) \times \max(0, \text{margin} - D)^2$, where D is the Euclidean distance between embeddings of two samples, $y = 1$ for same-patient pairs and $y = 0$ for different-patient pairs, and margin is a hyperparameter.

Rationale: Contrastive learning explicitly optimizes the embedding space so that same-patient samples cluster together while different-patient samples are pushed apart. This formulation is well-suited for verification tasks and naturally handles the extreme class imbalance through pair sampling strategies. The shallow architecture is adopted to limit model capacity and reduce overfitting risk given the small sample size (336 samples).

2.2 Optimization process

All models are evaluated under patient-level 5-fold cross-validation, where all time points from each patient are assigned to the same fold, preventing identity leakage. Positive pairs consist of all within-patient sample combinations (e.g., for a patient with 7 time points, $C(7, 2) = 21$ pairs). Negative pairs are sampled from between-patient combinations.

Hyperparameter search:

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

- Model 1: PValue threshold (grid: 1, 2, 3, 5, 10); minimum sample detection rate for SAAP inclusion (grid: 5%, 10%, 20%, 30%)
- Model 2: Same as Model 1, plus epsilon for frequency weighting (grid: 1e-6, 1e-4, 1e-2)
- Model 3: Number of PCA components (20, 50, 100), similarity metric (cosine, Euclidean), normalization method (none, L2), number of neighbors k (1, 3, 5, 10), 1.0, 2.0), dropout rate (0.3, 0.5), weight decay (1e-4, 1e-3), number of PCA components used as input (20, 50, 100)
- Model 4: Number of trees (100, 300, 500), max depth (5, 10, 20, None), min samples per leaf (2, 5, 10), number of features after initial filtering (top 500, 1000, 2000 by variance)
- Model 5: Embedding dimension (32, 64, 128), learning rate (1e-4, 1e-3), margin (0.5, 1.0, 2.0), dropout rate (0.3, 0.5), weight decay (1e-4, 1e-3), number of PCA components used as input (20, 50, 100)

The extreme feature-to-sample ratio (up to 40,921 features for 336 samples) makes overfitting the primary concern, so overfitting mitigation is necessary:

- Feature filtering and PCA are applied before model training to aggressively reduce dimensionality.
- Models 1 and 2 are non-parametric and cannot overfit in the traditional sense.
- Model 3 is non-parametric and less prone to overfitting. However, performance may degrade in high-dimensional space. Dimensionality reduction and normalization are therefore critical to ensure stable nearest-neighbor relationships.
- Model 4 can use out-of-bag error estimates, restricted tree depth, and minimum leaf size to limit complexity. We will monitor the gap between training and out-of-bag accuracy.
- Model 5 can use dropout regularization, L2 weight decay, early stopping (patience = 10 epochs on validation loss), and a deliberately shallow architecture. Batch normalization is applied to stabilize training with small batches.
- For all models, we will report both training and validation performance and flag any model with a training-validation gap exceeding 0.1 AUROC as potentially overfit.

2.3 Feature selection

Feature selection is essential given the extreme dimensionality and sparsity of the data. We employ an approach consisting of the following steps:

1. Missingness-based filtering

Remove peptidoforms/variants detected in fewer than a threshold percentage of samples. Preliminary analysis shows that 38,447 of 40,921 peptidoforms (94.0%) have over 50% missing values. Setting a threshold at 50% maximum missingness retains 2,474 peptidoforms; at 20%, approximately 1,500 features remain. For the Variants intensity table, 4,003 variants have less than 50% missing values, and 2,002 are detected in all weight-loss samples. We will experiment with thresholds of 20%, 30%, and 50%.

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

2. Variance-based filtering

Among retained features, remove those with near-zero variance across samples (bottom 10th percentile of log-intensity variance), as constant or near-constant features cannot distinguish between patients.

3. Annotation-based selection for SAAP models

For Models 1 and 2, restrict features to the 6,956 SAAP-annotated peptidoforms. After applying the missingness filter (50% threshold), approximately 266 SAAPs remain. SAAPs directly reflect genotype and are the theoretically most stable identity markers. Further refinement uses PValue filtering to retain only high-confidence SAAP identifications.

4. Model-driven selection

After initial model training with Model 5, extract the top features by Gini importance or permutation importance. Retrain a reduced model on the top 50, 100, and 200 features and compare performance to the full model. This will reveal whether a small set of features suffices for identification and will prioritize specific peptidoforms for the Stage 2 critical validation.

3. Evaluation of the project models

3.1 Data division into training, validation, and test sets

Since this dataset contains repeated blood samples collected from the same patient across multiple time points, the data should be split at the patient level rather than at the sample level. This follows from the fact that longitudinal repeated measures are correlated within subject, and sample level splitting can introduce subject leakage or identity confounding, leading to overly optimistic estimates of model performance[1][2]. A reasonable strategy is to assign approximately 60% of patients to the training set, 20% to the validation set, and 20% to the test set, while keeping all samples from each patient within a single split. Because the cohort is relatively small, an alternative and likely more robust strategy is **patient level k fold cross validation**, which would make better use of the available data and provide more stable estimates of model performance. The external COVID19 sera dataset should not be mixed into the main longitudinal identity splits as positive examples. Instead, it should be used as an **independent negative reference set** during validation and testing to measure whether the model incorrectly matches samples from unrelated individuals across datasets.

3.2 Scoring metrics and rationale

The most important goal of this project is to determine whether the model can correctly identify samples from the **same patient across different time points**, while avoiding incorrect matches

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

between **different patients**, especially between this weight loss cohort and the external COVID19 cohort. For this reason, evaluation should include both **pairwise matching metrics** and **retrieval style metrics**. For pairwise evaluation, useful metrics include **AUROC**, **AUPRC**, and **true positive rate at a fixed low false positive rate**. AUROC measures overall discrimination between same patient and different patient pairs, while AUPRC is particularly useful when the positive class is relatively rare. True positive rate at low false positive rate is especially important here because in an identifiability setting, even a small number of false matches can be problematic. For retrieval based evaluation, metrics such as **Recall@1**, **Recall@k**, and **mean average precision** are appropriate, because they test whether a query sample ranks its true same patient matches above all other candidates. Together, these metrics capture the key qualities of a good model: longitudinal consistency within an individual, strong separation between individuals, and robustness against spurious cross dataset matches.

3.3 Impact of data imbalance and reference baselines

Imbalance is expected to be an important issue in this setting, because the number of negative comparisons between different patients will greatly exceed the number of positive comparisons between samples from the same patient. This imbalance becomes even more pronounced when unrelated samples from the external COVID19 sera dataset are included as additional negatives. For this reason, overall accuracy alone will not be considered an adequate measure of model quality, as it may overstate performance for models that simply predict the majority class. To provide meaningful context for evaluation, we will compare against several reference baselines, including a majority class baseline that predicts all pairs as different patients, a random scoring baseline, and simple similarity based baselines such as cosine similarity, Euclidean distance, or nearest neighbor matching after standard preprocessing. These baselines will provide a transparent point of comparison for determining whether the proposed approach captures patient specific biological signals beyond trivial or noninformative decision rules. A successful model should substantially outperform these baselines while maintaining strong discrimination, high retrieval quality, and low false positive matching rates under class imbalance.

4. Assessing limitations and alternative approaches

4.1 Expected limitations of the proposed approach

The proposed approach may be affected by several limitations of the data. Although the dataset contains 336 samples, these samples come from only 58 patients, and repeated samples across time points are not independent, so the effective sample size is still relatively small. This increases the risk of overfitting, especially for more flexible supervised models such as contrastive metric learning. In addition, the data are extremely high-dimensional and contain substantial missingness, which makes it difficult to distinguish informative identity-related features from noisy or unstable ones. Missingness-based filtering is necessary to reduce dimensionality, but it may also remove rare SAAPs that are useful for patient identification.

Sample Identifiability – Project Plan

Xuchao Zhou Haoxi Wu Naicheng Yu Yihang Chen Yujie Zhao

Another concern is that samples from the same patient may still change across time points, and the weight-loss dataset and the external COVID-19 dataset may also differ because they come from different studies. These differences may introduce variation that is not directly related to patient identity. Finally, because the number of negative pairs is much larger than the number of positive pairs, class imbalance may make both training and evaluation more difficult.

4.2 Possible alternative approaches to reduce these limitations

To reduce the impact of these limitations, the project can place greater emphasis on simpler and more interpretable methods if the more complex supervised models show unstable performance. In particular, approaches that rely on biologically grounded signals may be more robust because they are biologically motivated, require less training data, and focus directly on variant patterns that should remain stable across samples from the same patient. Simpler matching methods, such as Spearman's correlation, may also generalize better than a neural embedding model in a small-sample setting. In addition, the project could consider a missingness-aware representation by including feature detection indicators instead of relying only on observed abundance values, which may reduce bias caused by comparing only shared non-missing features. For the worldwide identifiability analysis, a more conservative alternative is to restrict the calculation to high-confidence SAAPs with available population-frequency information, which would make the results easier to interpret and more reliable.

Reference:

[1] Schober and Vetter, 2018, *Repeated Measures Designs and Analysis of Longitudinal Data*

[2] Garcia et al., 2017, *Statistical Approaches to Longitudinal Data Analysis in Neuroimaging*